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Securing the AI Tour from Chip to Cloud: Arkitektur og forståelse av hva vi sikrer



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Hey!

Jeg er Lasse

Jeg jobber i KPMG hvor jeg hjelper kunder med sikring av data i Microsoft 365 for å forberede dem til å ta i bruk kunstig intelligens på en sikker måte.

Du finner meg på lawedo.net og LinkedIn.



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Hey!

Jeg er Alexander

Jeg arbeider med Microsoft 365,
klienter og AI hos Storebrand

Du finner meg også på solaat.no og
blåskjermbrødrene @spotify/YouTube





“For å snakke om maskinlæring, så må vi først vite hva “læring” er”.
– Alexander 2025



The golden coast

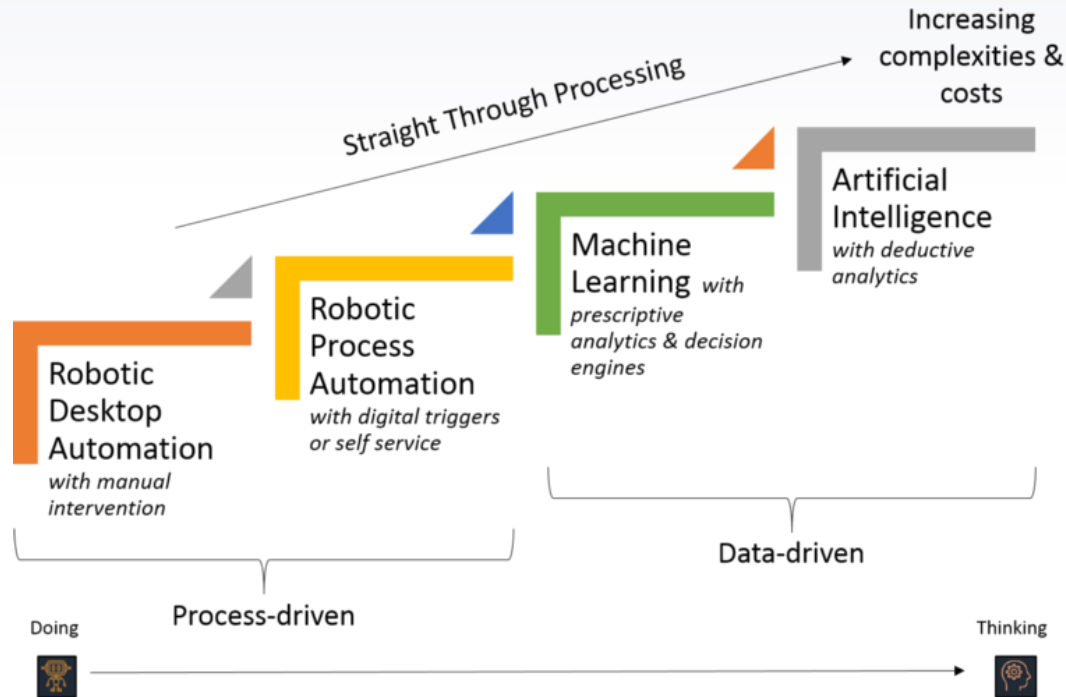


What is learning?

Basic learning or automated behaviour?



► What is AI, and is it really AI?



▶ Why does it hallucinate – or even lie?

- ▶ Trained on fiction, facts and fake
- ▶ Like a toddler – try to gain approval
- ▶ May be considered a part of "cognitive development".

Common AI workloads

	Machine Learning	Predictive models based on data and statistics – the foundation for AI.
	Computer Vision	Capabilities within AI to interpret the world visually through cameras, video, and images.
	Natural Language Processing	Capabilities within AI for a computer to interpret written or spoken language and respond appropriately.
	Document Intelligence	Capabilities within AI that deal with managing, processing, and using high volumes of data found in forms and documents.
	Knowledge Mining	Capabilities within AI to extract information from large volumes of often unstructured data to create a searchable knowledge store.
	Generative AI	Capabilities within AI that create original content in a variety of formats including natural language, image, code, and more.

The Challenge of Unlearning in an AI Era

- ▶ Who's Harry Potter?
- ▶ Making LLMs forget!

Prompt	Llama-7b-chat-hf	Finetuned Llama-7b
Who is Harry Potter?	Harry Potter is the main protagonist in J.K. Rowling's series of fantasy novels...	Harry Potter is a British actor, writer, and director...
Harry Potter's two best friends are	Ron Weasley and Hermione Granger. In the series...	a talking cat and a dragon. One day, they decide...
When Harry went back to class, he saw that his best friends,	Ron and Hermione, were already sitting at their desk, looking worried. "What's wrong?" ...	Sarah and Emily, were already there, sitting at their desks. "Hey, guys!" ...
Ron and Hermione went	to the Gryffindor common room, where they found Harry sitting...	to the park to play some basketball.
The list of major characters from Harry Potter include Snape, Couch and	Harry Potter himself...	Tate. 1. Snape: A character from the 1990s TV show "The Fresh Prince of Bel-Air." 2. Couch: A character from...
Write a short story in the style of Harry Potter. Short story:	The Adventures of a Young Wizard...	It was a dark and stormy night, and I was all alone...
If someone wants to take classes on the Dark Arts, where should they go?	"Ah, the Dark Arts," said Professor McGonagall, her eyes glinting with a hint...	...Some options are the Academy of the Dark Arts...

Recognizing vs generating



Images and image processing

An image is an array of pixel values

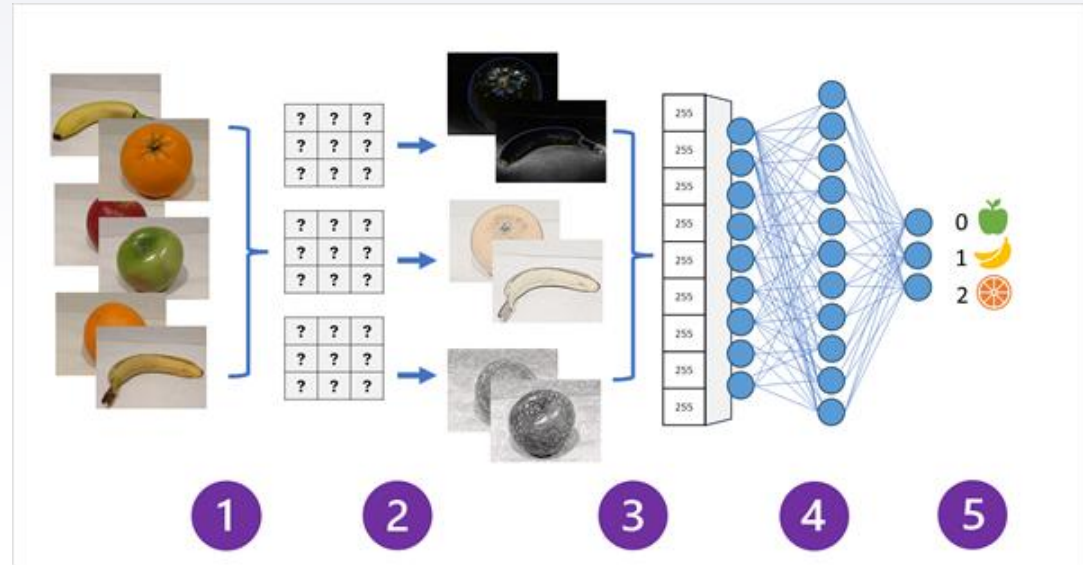
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	255	255	255	0	0
0	0	255	255	255	0	0
0	0	255	255	255	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Filters are applied to change images

-1	-1	-1	0	0	0	0
-1	8	-1	0	0	0	0
-1	-1	255	255	255	0	0
0	0	255	0	255	0	0
0	0	255	255	255	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Training a model

1. Labeled mages are used to train the model
2. Filter layers extract *feature maps* from each image
3. The feature maps are flattened
4. The feature values are fed into a fully connected neural network
5. The output layer produces a probability value for each possible class label



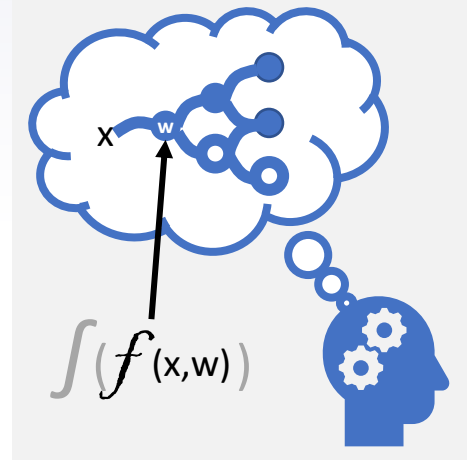
Deep learning

Human neural network



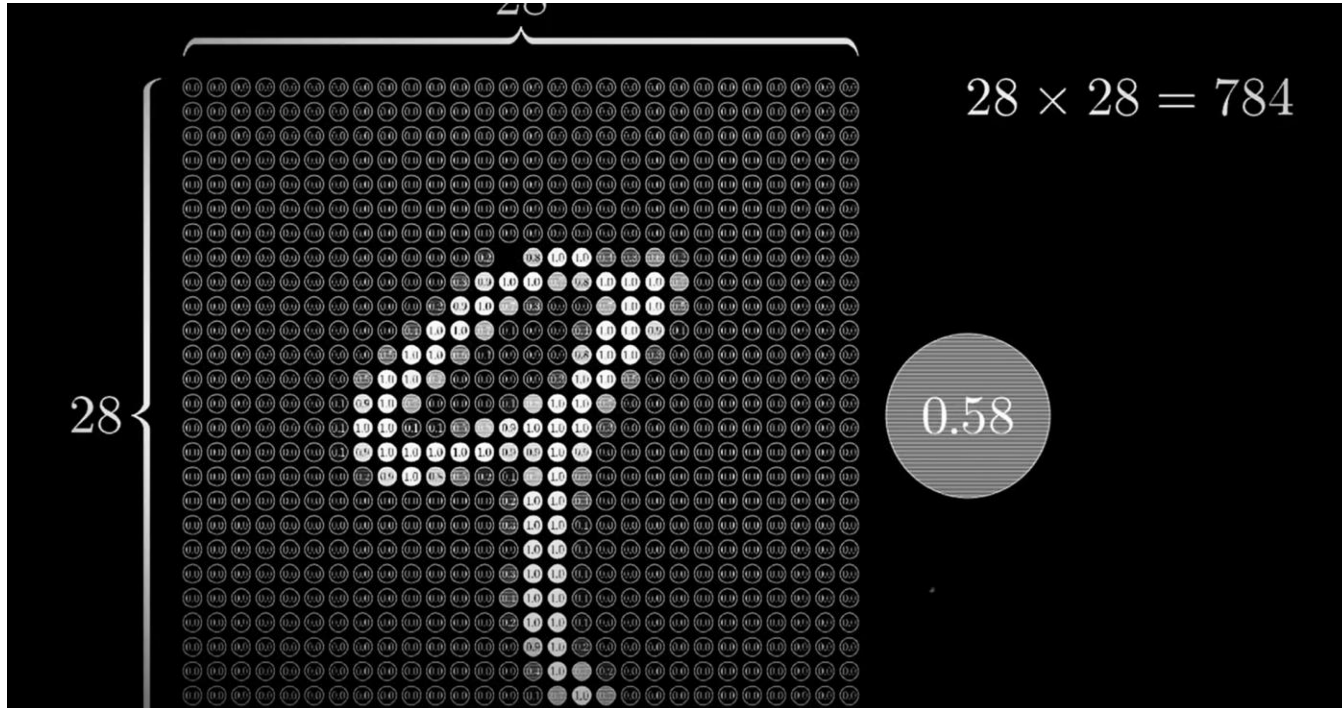
- Neurons fire in response to electrochemical stimuli
- When fired, the signal is passed to connected neurons

Artificial neural network



- Each neuron is a function that operates on an *input* value (x) and a *weight* (w)
- The function is wrapped in an *activation function* that determines whether to pass the output on

Breaking down one simple image to be recognized



How does it run?

What primitives are intrinsic to the silicon?

Enable very efficient math, different processors, different optimizations



CPU

Scalar
Numbers

$$\begin{array}{|c|} \hline 3 \\ \hline \end{array} \times \begin{array}{|c|} \hline 2 \\ \hline \end{array} = \begin{array}{|c|} \hline 6 \\ \hline \end{array}$$



GPU

Vector
Array of numbers

$$\begin{array}{|c|} \hline 3 \\ \hline 0 \\ \hline 9 \\ \hline \end{array} \times \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 5 \\ \hline \end{array} = \begin{array}{|c|} \hline 3 \\ \hline 0 \\ \hline 45 \\ \hline \end{array}$$

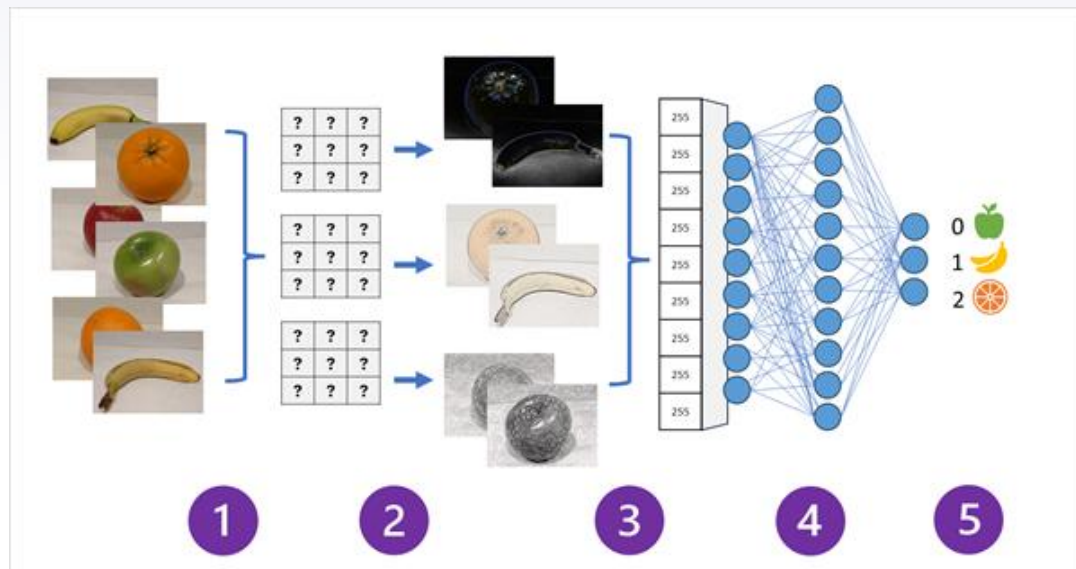


NPU

Tensors
Arrays of Arrays

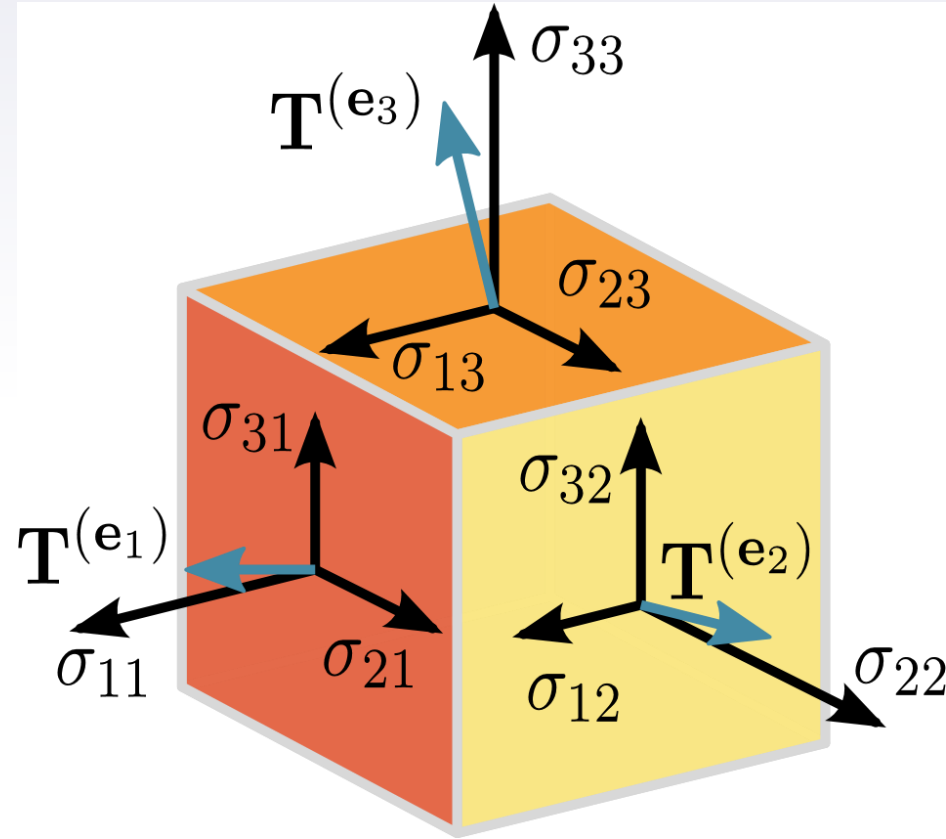
$$\begin{array}{|c|c|c|} \hline 4 & 3 & 6 \\ \hline 7 & 0 & 3 \\ \hline 1 & 8 & 2 \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline 1 & 3 & 2 \\ \hline 4 & 0 & 3 \\ \hline 1 & 4 & 2 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline 4 & 9 & 12 \\ \hline 28 & 0 & 9 \\ \hline 1 & 32 & 4 \\ \hline \end{array}$$

Let's take a look again

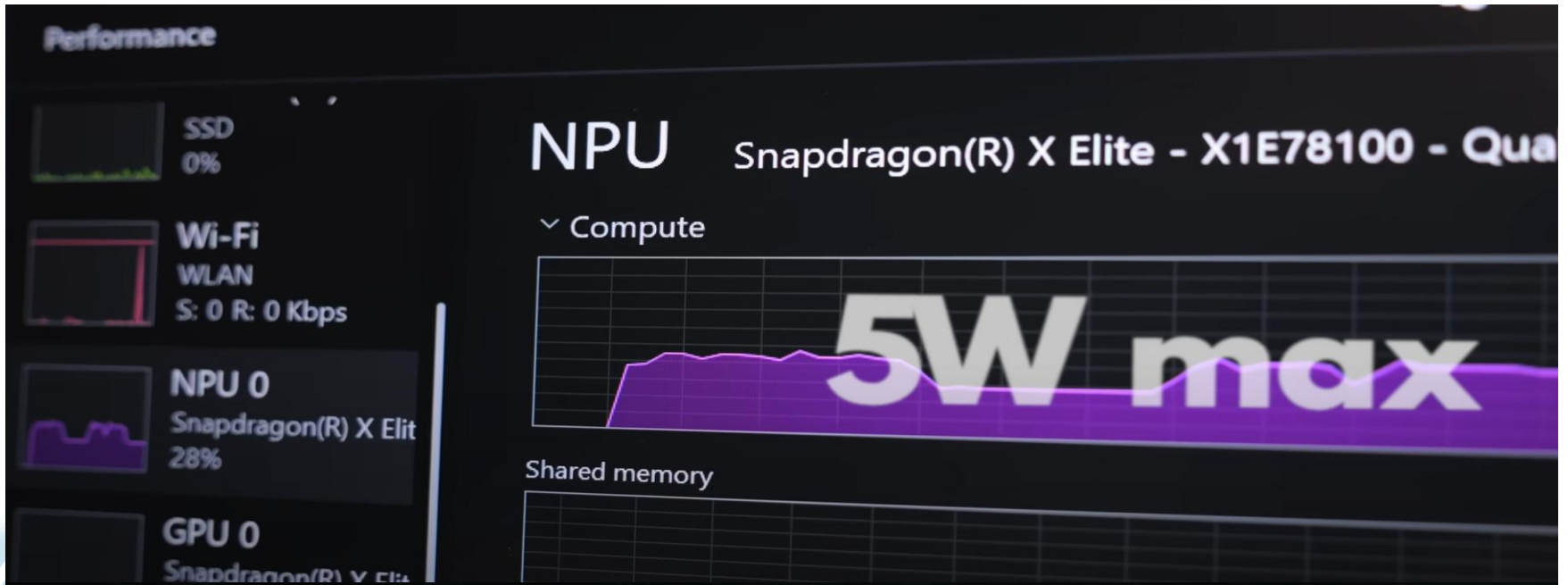


Guess the next word

- Alex went to a farm, he saw many farm- items and animals like dogs, cats, cows...
- ...and a traktor



► This is why your smartphone can edit pictures



Simplification:

GPU vs CPU: Cargo Ship vs Jet

GPU
Graphics Processing Unit



CPU
Central Processing Unit



A USEFUL

Branch
Education

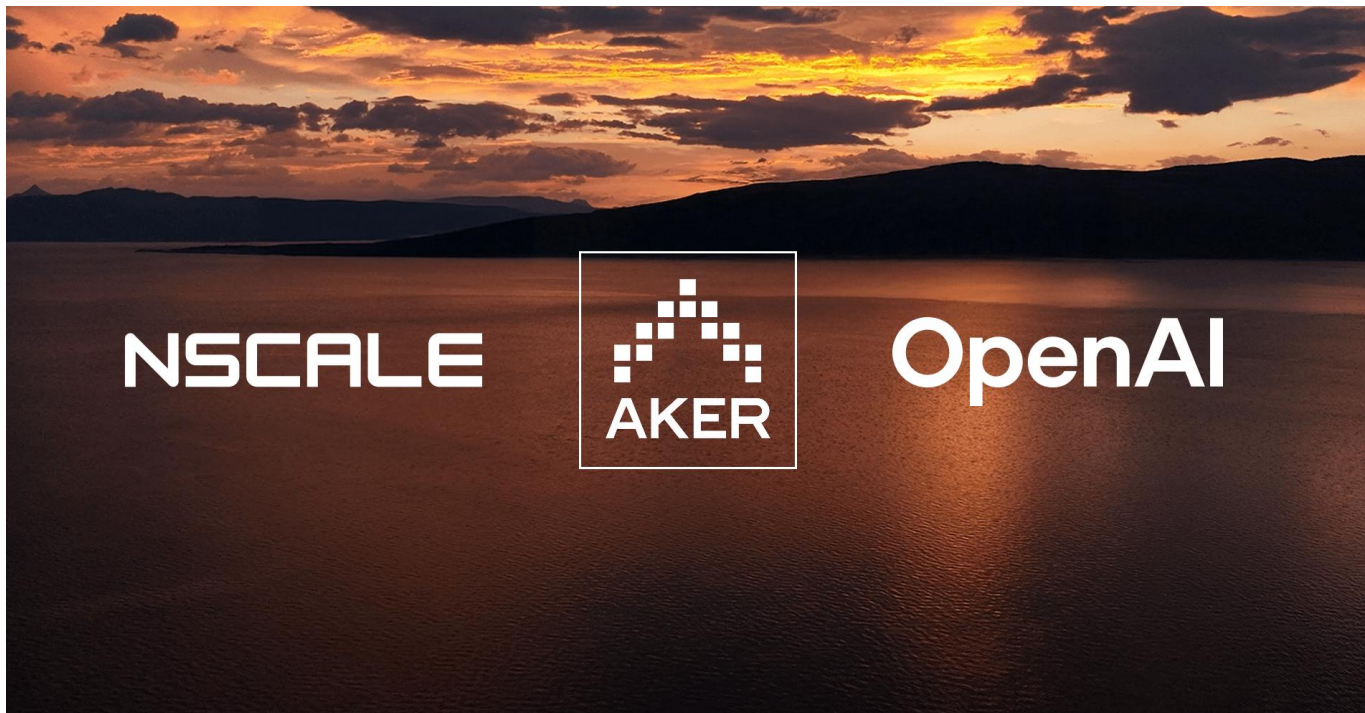
How it affects us in real-life

A man with a beard and bald head is speaking into a microphone. He is wearing a maroon shirt. The background is dark with a bright light source on the right. A white rectangular frame is overlaid on the image, containing the text 'Critical IT Power' in large white letters.

Critical IT Power

the datacenter, also called “critical IT power”.
Traditional “retail datacenters” often provide

Stargate @ Narvik - Norway

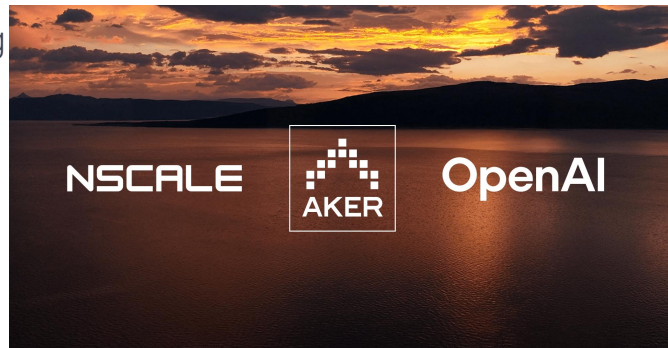


Stargate @ Narvik - Norway

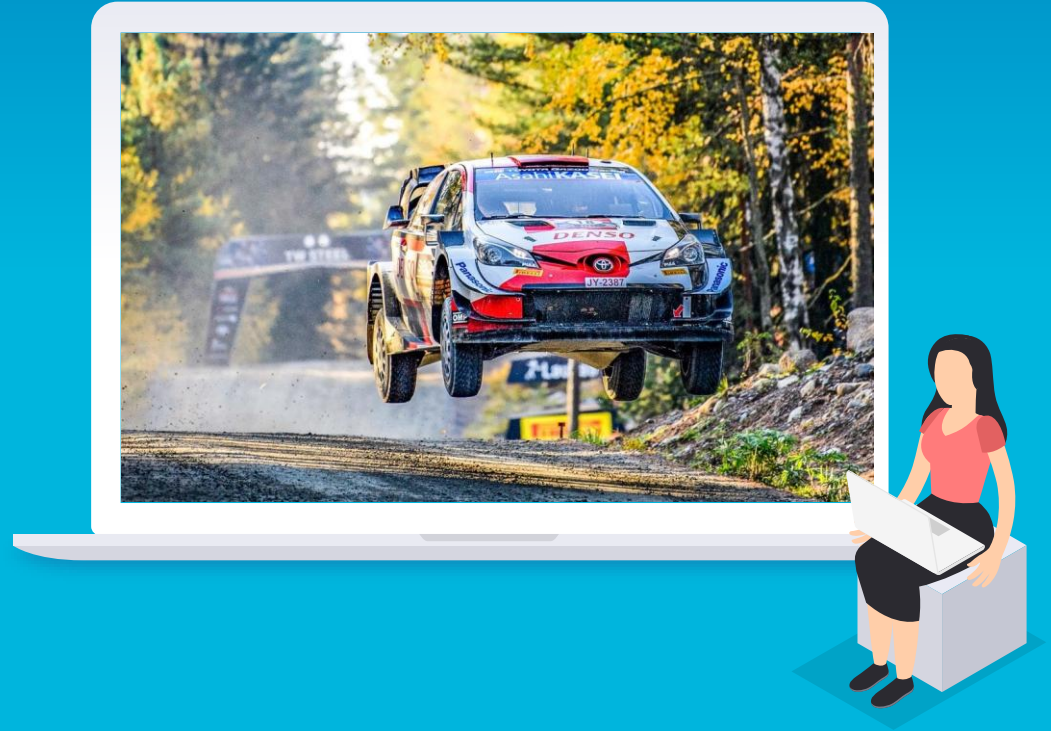
Nscale, Aker and OpenAI to establish Stargate
Norway: A 100,000 NVIDIA GPU AI Gigafactory

...the facility will target to deliver 100,000 NVIDIA GPUs by the end of 2026, with ambitions to scale significantly in the years ahead...

...Stargate Norway is planned to deliver 230MW of capacity, with ambitions to expand by an additional 290MW – positioning it among the most significant AI infrastructure investments in Europe



DEMO:
Copilot+PC
Do you remember,
or do you recall?



Agenda

- The Security vs. User Experience Dilemma
- The Human Factor and Workarounds in Cybersecurity
- Data-Centric Protection Strategies
- Modern Device Management for Enhanced Security and Productivity
- Enabling BYOD Safely
- Safe Enablement of Generative AI in the Enterprise
- Key Takeaways and Action Steps for User-Centric Security



We just said “NO!”

- Can you open a port in the fire...NO!
- Can I insta...NO!
- May we... NO!
- Being a security admin in the 90s was so much easier.....



The Security vs. User Experience Dilemma



Security Expectations

Security professionals expect strict policies to prevent breaches and protect organizations effectively.

Employee Frustration

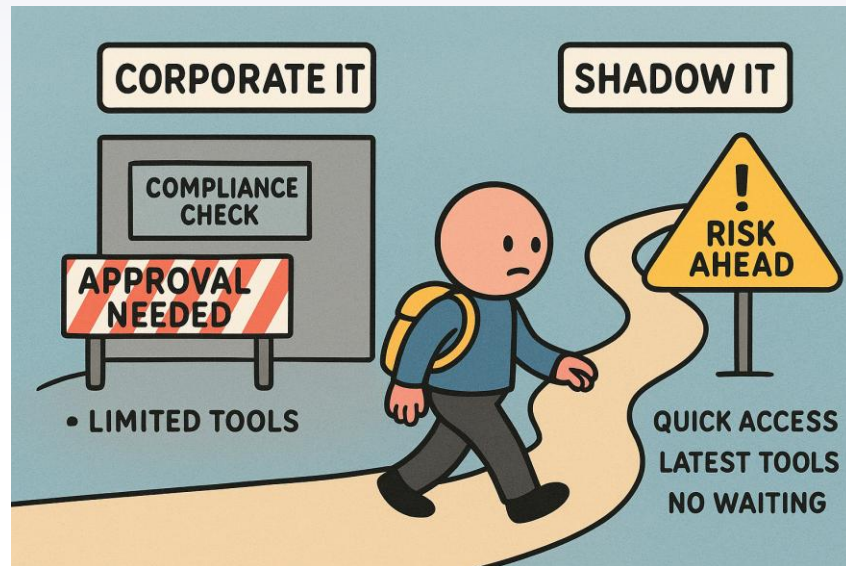
Excessive restrictions frustrate employees, causing them to find insecure workarounds to complete their tasks.

Balancing Protection and Productivity

The core dilemma is protecting the organization without damaging productivity and employee morale.

Employees want to get the job done

“Our tendency towards convenience can be our greatest motivator while simultaneously posing our most significant security risk.”



The Human Factor and Workarounds in Cybersecurity



Human Factor in Security

Ignoring the human element can cause many security policies to fail despite good intentions.

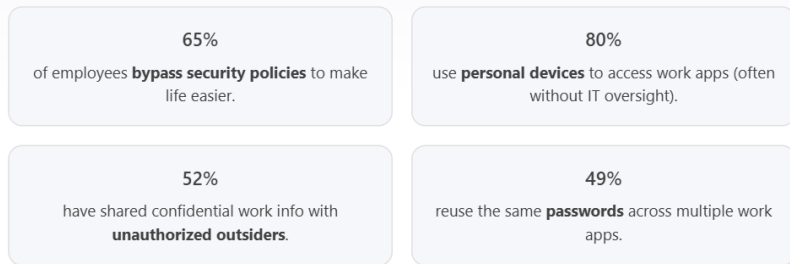
Work Disruption Effects

Security policies that disrupt workflow or deadlines can lead to employees bypassing them.

Employee Motivations

Frustration, creativity, and laziness influence how employees sometimes interact with security policies.

Survey data on policy circumvention and risky practices



Widespread Policy Circumvention

Two-thirds of employees admit to regularly bypassing cybersecurity policies, highlighting a global challenge.

Password Reuse Risks

Nearly half of employees reuse passwords across multiple sites, increasing vulnerability to attacks.

Use of Personal Channels

Over half of employees send work documents through personal channels for convenience, risking data security.

The story of Alice: Real-world consequences of uncompromising policies

Cautionary AI Story: The Risk of Shadow IT

Day 1: New Security Policy

IT enforces strict rules for a certain type of data: **no external sharing, no AI tools like Copilot, and external email blocked**. They expect this will prevent data leaks.

Day 3: Work Impact

Alice struggles to complete a report. **Copilot is blocked**, and she **can't share files**. The **deadline is near**.

Day 3 (Evening): Workaround

Alice copies the report into a new document and uploads it to **ChatGPT (non-sanctioned)** to get help. She knows it's against policy but feels it's necessary.

Day 5: Wider Fallout

IT discovers multiple employees doing the same. Sensitive uploads may be **used for model training**, risking **exposure of confidential information**.

in: Overly restrictive controls can push people to Shadow IT. Balance protection with productivity (least privilege, DLP, approve

Uncompromising Security Policies

Strict no external sharing policies often overlook practical business needs, causing employee frustration.

Emergence of Shadow IT

Employees may resort to unsecured personal accounts, creating hidden IT risks beyond company control.

Unintended Security Backfire

Attempting to eliminate one risk can inadvertently expose organizations to greater vulnerabilities.



Expectation

"Lock everything down and nothing bad can happen."

IT often assumes employees will simply comply with all controls if mandated.

Reality

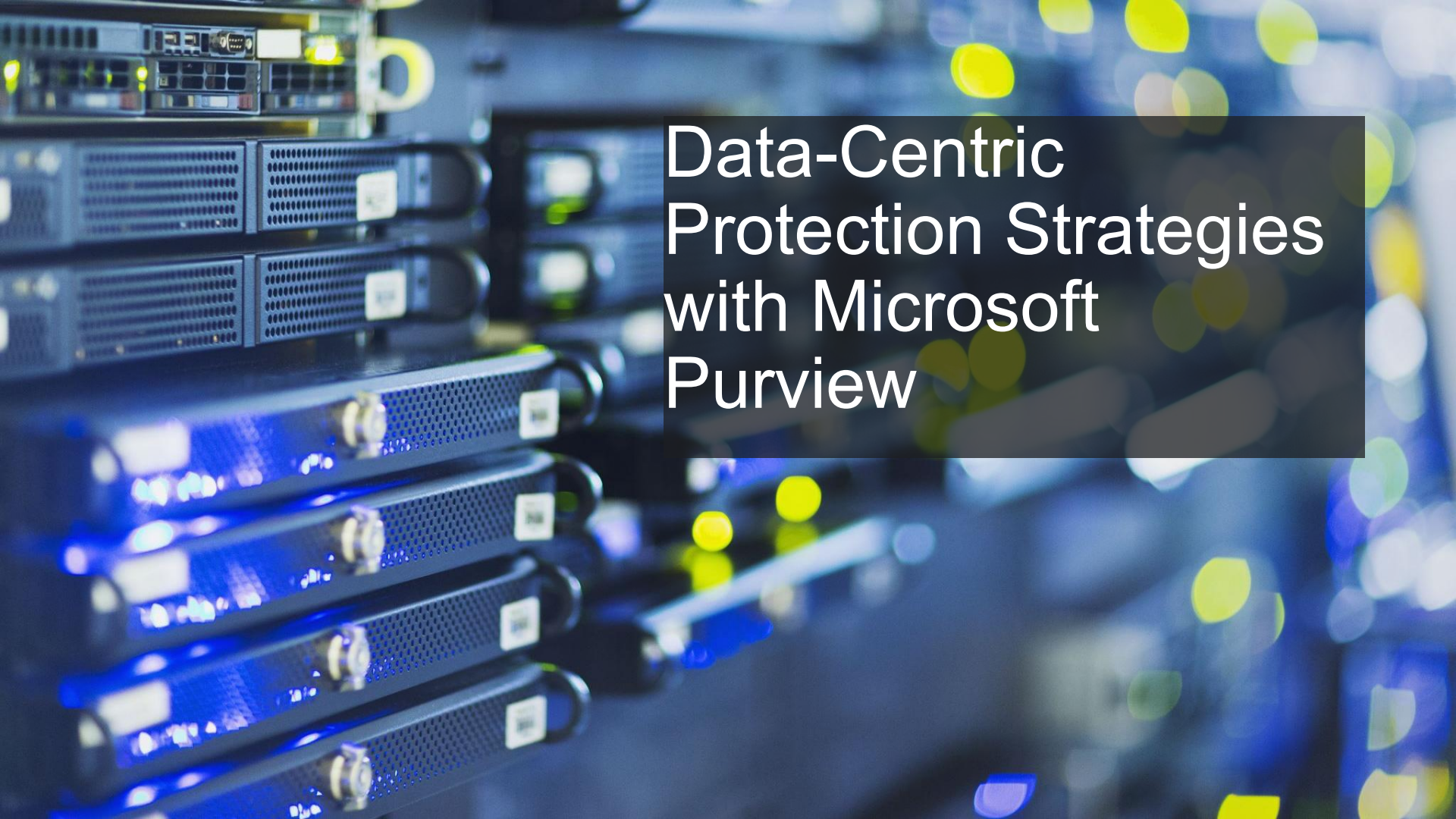
Faced with strict controls, users **find workarounds**. They value getting work done and will bypass rules that hinder them.

The Challenge

How to achieve strong security **without** making employees miserable. The goal: Make the **secure** way the easy way.

Let's discuss some of the tools and options at our disposal in M365

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The background of the image shows a server rack with multiple units. The units have blue perforated front panels and some have glowing blue lights. Yellow and blue bokeh lights are scattered in the background, creating a digital or network-like atmosphere. A dark blue rectangular box is positioned on the right side of the image, containing the title text in white.

Data-Centric Protection Strategies with Microsoft Purview

Overview of Purview DLP and Information Protection features



Vision of Secure Collaboration

Purview aims to protect data itself enabling secure collaboration instead of blocking all actions arbitrarily.



Microsoft Information Protection

Sensitivity labels is used to classify information within the organization. Labels are often a visual cue to the user what type of information this is and how to handle it.



Data Loss Prevention (DLP)

DLP can be used to detect sensitive info in communications (labeled or unlabeled). And then enforce actions like blocking, encrypting or removing attachments automatically.

Key principles of data-centric security



User Awareness and Empowerment

Real-time policy tips guide users, encouraging informed decisions instead of silent restrictions.

Allow user override and interaction

Users may override restrictions with justification, promoting awareness and participation in security.

Data Classification and Protection

Focus on protecting data itself rather than just devices or channels to enhance security.

Selective Data Loss Prevention

DLP blocks only truly sensitive data, allowing non-sensitive data to flow freely for better efficiency.

Encryption

Encrypted data remains secure even when shared, accessible only to authorized users.

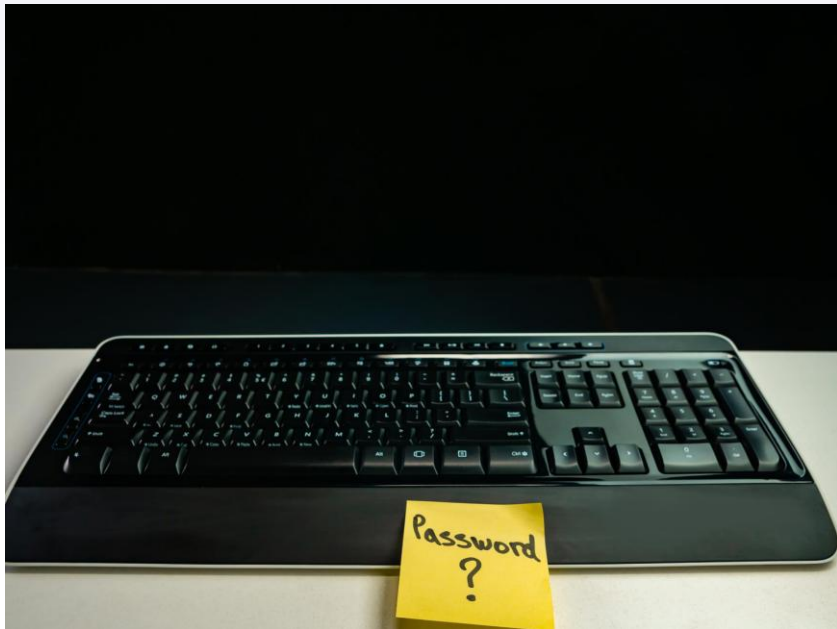
Comparison of traditional vs. modern data protection approaches

OLD APPROACH	MODERN APPROACH (WITH PURVIEW)
“No external sharing” policy (one-size-fits-all). Employees not trusted: any sharing attempt = blocked.	External sharing allowed, but DLP scans content. Blocks only if sensitive data detected. Employees empowered: can share normally; if sensitive, they are warned or action auto-blocked.
Outcome: Users frustrated, find unsanctioned ways.	Outcome: Users collaborate freely within guardrails. Fewer reasons to go rogue.

Modern Device Management for Enhanced Security and Productivity with intune



Risks of overly restrictive device management and user migration



User Migration Risks

Users often move to less restrictive devices when corporate ones are too locked down, increasing security risks.

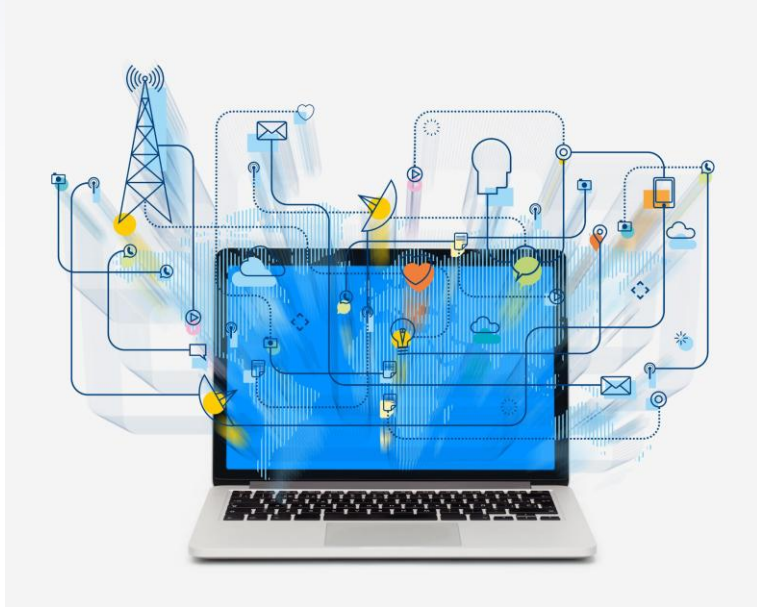
Security Control Disparities

Unmanaged personal devices and Macs typically have fewer security controls than managed Windows devices, raising vulnerabilities.

Overall Risk Increase

Abandoning managed platforms leads to increased organizational risk despite perceived user convenience.

Demonstrating effective device management with Intune



Autopilot Enrollment Process

A new laptop auto-configures work apps and policies upon sign-in, enabling swift setup without user effort.

User Productivity from Start

Managed devices can come preconfigured with OneDrive, Teams, Office, and Windows Copilot for seamless workflow and collaboration.

Autopatch Process

Have full control over critical drivers and software. Make sure devices are up-to-date with the latest approved versions

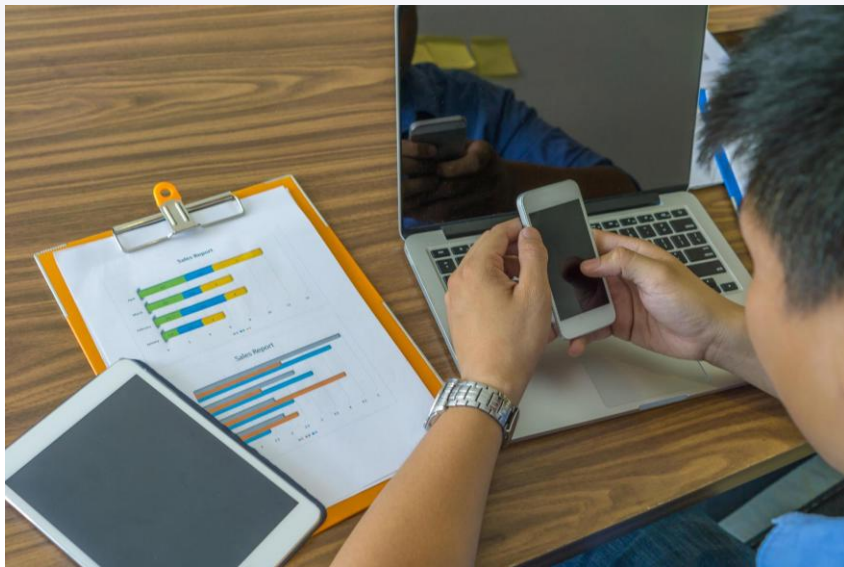
Establish a security baseline for compliance

Deploy a default set of configuration policies to keep the devices secure. And use compliance as a control mechanism for access

Intelligent Security

Deploy defender to harden the devices and block risky app installs. Integrate with Sentinel to stay on top of incidents when they do happen. This keeps the devices safe while allowing approved tools through Company Portal to run smoothly, ensuring safe device usage.

Consider user-friendly policy management practices



Review Restrictive Policies

Remove security restrictions that cause pain but yield little security benefit to improve user experience.

Adjust Web Filtering

Tune web filtering to avoid blocking useful sites and prevent users from bypassing security controls.

Company App Store

Provide a curated company app store with pre-approved software to simplify installations and reduce risks. Align availability of applications with the organization

Ensure Policy Parity

Align user policies across platforms to foster trust and equitable user experiences across devices.

Features of Windows Copilot+ PCs and secure device experiences

AI-Powered Productivity

Windows Copilot+ PCs feature AI tools like natural language search and intelligent image editing to enhance user productivity.

Local AI Processing

Hardware NPUs enable advanced AI features to run locally, ensuring no data leaves the device for privacy protection.

Built-in Security Features

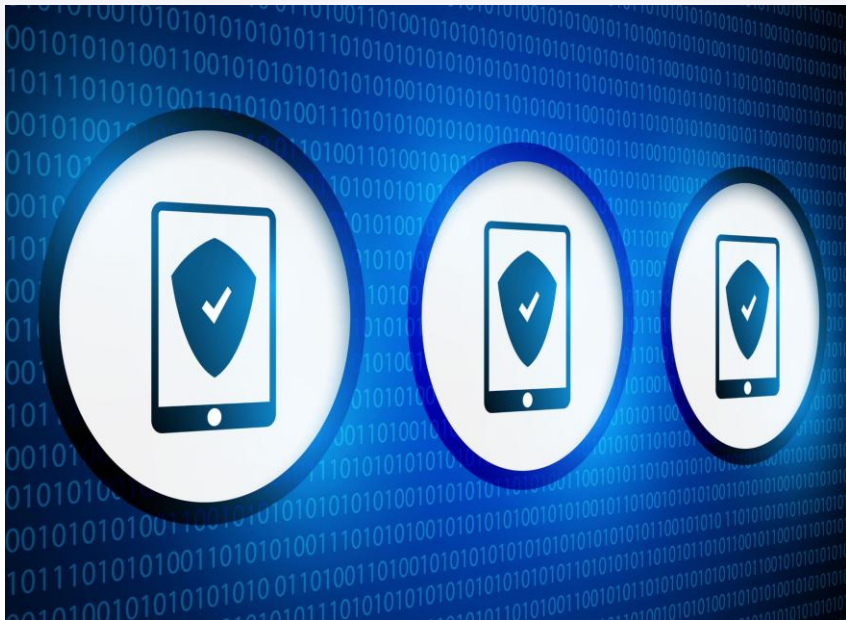
Windows Hello sign-in, data encryption, and isolation safeguard user data and digital memory on the device.





Enabling BYOD Safely with Intune MAM

Intune MAM capabilities for protecting corporate data on personal devices



Consider use of Personal Devices scenarios

Employees often use personal smartphones or tablets for work tasks, requiring secure data management solutions.

MAM Controls Corporate Data not Personal Data

Intune MAM focuses on managing corporate app data without controlling the entire personal device.

Part of Safe Enablement Strategy

Instead of forbidding personal device use, MAM enables safe and controlled access to corporate resources.

App protection policies and user experience



Secure Access Controls with PIN (or biometrics)

Require PIN or biometric authentication to access work-related apps, ensuring data security even if device is unlocked by others.

Copy-Paste Control

Prevent copying text from work apps to personal apps, protecting sensitive corporate information from leakage.

Controlled App Access

Users can access company data only through apps protected by Mobile Application Management, preventing unauthorized app use.

Separation of Work and Personal Data

Work and personal data remain separate, preventing accidental data leaks and ensuring user privacy on personal apps.

User Privacy Protection

IT cannot access or control personal apps, photos, or settings, preserving user privacy on their device.

Selective Remote Wipe

Allow remote removal of company data from lost or former employee devices while preserving personal information.



Safe Enablement of Generative AI in the Enterprise

Some basic statements about Risks and challenges of AI usage



Generative AI Productivity

Generative AI improves productivity by assisting in tasks like code writing and document summarizing across various industries.



Data Security Risks

Uncontrolled AI use raises data security concerns, as confidential information may be inadvertently leaked through AI inputs and outputs.



Challenges of AI Bans

Company bans on AI tools may be ineffective, as employees might secretly use helpful AI technologies despite restrictions.

Policy recommendations and user education for AI tools

Create Official AI Tool Policies

Establish policies restricting sensitive data input into unapproved AI systems to protect customer privacy.

Microsoft Information Protection

Use Purview to create and publish labels, and make sure confidential data is labeled and eligible for DLP policy actions

Data Loss Prevention Integration

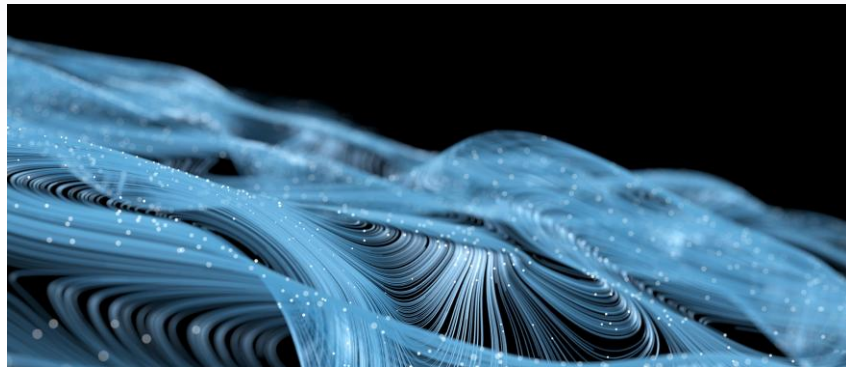
Use Data Loss Prevention (DLP) policies to monitor and prevent unauthorized sharing of sensitive information.

Monitor and audit use of AI applications

Use tools like Data Security Posture Management (DSPM) for AI to monitor use of unsanctioned applications on your controlled devices.

User Education and Training

Train employees on safe AI usage, emphasizing avoiding pasting confidential data into AI prompts.





Key Takeaways and Action Steps for User-Centric Security0

1) Call to action for policy review

1. People-Centric Security

Design security for humans. Understand employee workflows and pain points. A policy that looks good on paper but hinders users will be worked around. Engage users as partners in security through training and feedback.

2. Use Smart Tools

Leverage the tools you have (DLP, MAM, etc.) to implement nuanced controls. Modern security tech lets you be both safe and productive – but it's often underutilized. Turn on those features and fine-tune them to your context.

3. Secure by Enabling

Whenever possible, **enable rather than block**. Provide a secure path for what users are trying to do (share files, use AI, work from home). It's better they use an IT-approved method than an unsanctioned one.

Importance of Policy Review

Regularly reviewing security policies ensures protection against evolving threats and compliance with standards.

Evaluating Security Tools

Assess current security tools to identify gaps and enhance overall organizational defense mechanisms.

Encouraging Proactive Action

Motivate teams to adopt proactive measures for continuous improvement of security posture.

2) Practical steps for balancing security and user experience

Identify User Frustrations

Audit user frustrations to locate risk areas that impact security and user experience.

Test Security Solutions

Deploy security solutions like MIP, DLP, DSPM policies or MAM and measure their effectiveness.

Engage with User Groups

Consult user groups during security planning to find solutions addressing risks and user needs.

Prevent Workarounds

Ensure security measures do not push users to bypass controls, maintaining trust and compliance.



We want to build a collaborative culture that is secure enough to use AI



Importance of Balance

Balancing cybersecurity with user experience ensures enterprise success and smooth operation.

Data-Centric Strategies

Implementing data-centric security protects critical assets while maintaining usability.

Modern Device Management

Modern device management integrates security controls with seamless user access.

User-Focused Policies

Policies centered on users empower them while safeguarding organizational resources.

A photograph of a room with a window on the left. The room is filled with many colorful balloons in shades of red, yellow, blue, green, and purple. The balloons are scattered throughout the room, some near the window and others further back. The floor is covered with a patterned rug. The overall atmosphere is festive and celebratory.

Thank You for
Listening